Kexuan (Michael) Huang

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Graduate Student

EDUCATION

University of Michigan 2022.08 - 2024.05
M.Sc. in Information Ann Arbor, MI, U.S.

University of Wisconsin-Madison 2022.01 - 2022.05
Exchange Student in Computer Science Madison, WI, U.S.

Shanghai Jiao Tong University 2018.09 - 2022.08
B.Eng. in Electrical and Computer Engineering Shanghai, China
Dual Minor in Computer Science and Entrepreneurship

INTERNSHIPS

UM-SJTU Joint Institute, SJTU 2021.05 - 2021.12
Teaching Assistant & Research Intern Shanghai, China

- Hold workshops, grade assignments & exams to help around 100 students in courses VE300 & VE496.
- Reproduce and optimize high-speed signal-free intersection models from papers in *SUMO* software with *Python*.

Advanced Micro Devices, Inc. (AMD) 2020.12 - 2021.05
Design Verification Engineer Intern, dGPU SOC, Shanghai R&D Center Shanghai, China

- Work on *HDL* code and waveform simulation to do RTL troubleshooting.
- Use *Python*, *Shell*, *Perl* & *Ruby* to code automation scripts and maintain regression flow. Use *Pandas*, *NumPy*, and *Sklearn* to analyze data, use *Requests and RegEx* to scrape website, and use *Django* to set up team webpage.

Segway-Ninebot, Inc. 2020.01 - 2020.03
Embedded Software Engineer Intern, IoT R&D Department. Shanghai, China

- Work on development of autonomous delivery robot, typically in communication & decision-making strategy optimization.
- Use *C* to do embedded programming with *STM* chip, prototype PCB verification and feedback reporting.

PROJECT EXPERIENCE

Big Data Analysis on Million Song Dataset 2022.06 - 2022.08
Course Project, UM-SJTU Joint Institute

- Compact and convert *hdf5* files to *Avro* files with *Python* scripts. Retrieve information from *Avro files* using *SQL* query in *Drill.*
- Analyze 20 GB dataset using parallelized BFS algorithm with *Hadoop MapReduce* and *Spark on clusters*.

Intelligent firefighting system based on Internet-of-Things technology, Frontend Team Leader 2021.07 - 2021.12
Capstone Project, UM-SJTU Joint Institute

- Develop an integrated IoT firefighting system to perform fire risk management with greater accuracy, better efficiency and lower cost using FAHP algorithm. The system is deployed on servers of Shanghai fire brigade currently.
- Implement backend with *FastAPI*, webpage with *React.js* and a cross-platform app with *React* Native. Use *MongoDB* as database, whose current data-handling quantity is at 7 million.

Digital Innovation of the Fever Thermometer, Team Leader 2020.09 - 2021.03
Entrepreneurial Project (Excellent: 3 in 26), Antai College of Economics & Management

- Develop a WeChat mini-program with *JavaScript*, *HTML* and *CSS*. Design the circuit and PCB of infrared thermometer, which can connect to mobiles via *Bluetooth*. Temperature registration can be done with greater efficiency in COVID-19 pandemic.

HONORS & AWARDS

Merit Undergraduate Scholarship, SJTU 2021.11
Advanced Teaching Assistant, UM-SJTU Joint Institute 2021.8
Most Improved Scholarship, SJTU 2020.10

SKILLS

- **Languages:** English (IELTS 7.5 / CET-6), Chinese (native)
- **Programming:** Assembly (x86), Verilog, C, C++, Java, Python, MATLAB, TypeScript, HTML, Shell, SQL, Markdown, LaTeX
- **Development:** Git, GitHub, Drone CI, Docker, Hadoop, Spark, Drill, Vim, VS Code, CCStudio, macOS, Linux
- **Hobbies:** Badminton (former professional athlete & SJTU team member), Anime Piano